

# Reading the Reader

## Adaptive Reading Systems: A Modular Software Architecture

Master Thesis







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Master Thesis

Date, year

By

Sachin Baral

Satish Gurung

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## **Abstract**

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

## Acknowledgements

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# **1 Introduction**

**1.1 Motivation and Context**

**1.2 Problem Statement**

**1.3 Research Questions and Objectives**

**1.4 Contributions**

**1.5 Scope and Delimitations**

**1.6 Thesis Outline**



## 1.7 Key Concepts and Related Works

### 1.7.1 Key Concepts for Adaptive Reading Analytics

This section defines the core concepts underlying the thesis. Definitions are grounded in primary and project-aligned sources.

#### **Adaptive Reading System**

The “Reading the Reader” concept assumes a closed-loop system that senses gaze-related signals (including pupil size) during reading, extracts higher-level events (fixations, saccades, regressions), classifies reading state or style, and triggers typographic interventions in real time [1].

#### **Micro-Interventions and Just-in-Time Adaptation**

The project proposal defines “micro-interventions” as small, context-preserving changes to typography or layout (e.g., spacing or contrast adjustments) designed to support reading without disrupting flow [2].

Within broader HCI literature, microintervention systems are described as structured, designed small actions embedded into systems, where timing, narrative coherence, and personalization influence overall user experience [2].

In this thesis, just-in-time adaptation is defined operationally as an intervention policy that triggers based on short temporal windows of online gaze and interaction features, while controlling for:

- Intervention cost (risk of disruption),
- Recovery behavior (time required to resume reading).

#### **Context Preservation and Recovery Metrics**

Recent ETRA 2025 work frames context preservation as interface functionality that enables readers to resume reading quickly after typographic changes. It introduces the metric Reading Resume Time (RRT) and compares four intervention designs: Popup, Undo, Notification, and Gradual [2].

Results indicate that a Gradual intervention combined with highlighting yields the fastest recovery time and highest user preference. This demonstrates that intervention form factor and transition smoothness significantly affect usability.

In this thesis, context preservation is treated as a first-class design goal with measurable outcomes:

- Reading Resume Time (RRT),
- Gaze stabilization time,
- Regression bursts following change.

### 1.7.2 Related Academic Work

#### **Context Preservation and Intervention Design**

The ETRA 2025 paper on context preservation formalizes a key HCI challenge: adaptive reading interfaces can either assist or hinder reading depending on how changes are introduced and how recovery is supported. In a within-subject study with 22 participants, significant differences in RRT were reported across intervention types [2].

This supports the thesis argument that intervention design is a measurable system behavior that must be architecturally supported through stable integration hooks and logging mechanisms.

### Reading Analytics for Emerging Content Types

An ETRA 2025 study comparing AI-generated and human-authored texts collected gaze data from 13 participants and applied I2MC for fixation identification. The study also analyzed pupil dilation and reading speed [2].

Findings demonstrate:

1. Gaze metrics are sensitive to subtle content differences,
2. Experimental reproducibility requires modular and transparent data pipelines.

### 1.7.3 Market Landscape of Tools and Products

#### Comparison of Tools

| Category                 | Vendor / Product           |     | Delivery Model      | Capabilities                                                                    | Pricing Signal            |
|--------------------------|----------------------------|-----|---------------------|---------------------------------------------------------------------------------|---------------------------|
| Screen-based eye tracker | Tobii Pro Fusion           |     | Hardware            | Up to 250 Hz sampling; fixation/saccade detection; pupil diameter recording [3] | Quote-based [3]           |
| Screen-based eye tracker | EyeLink 1000 Plus          |     | Hardware            | Binocular sampling up to 2000 Hz; research integration support [4]              | Quote request [5]         |
| Lower-cost eye tracker   | GazePoint HD               | GP3 | Hardware + Software | 150 Hz sampling; research-grade positioning [6]                                 | \$2,650 hardware-only [6] |
| Wearable eye tracking    | Pupil Labs Neon            |     | Hardware            | Academic pricing; complete research system [7]                                  | €5,515 academic [7]       |
| Remote webcam tracking   | RealEye                    |     | SaaS                | Subscription plans; data export; panel workflow [8]                             | Panelists \$15 [9]        |
| Experiment software      | Tobii Pro Lab              |     | Desktop software    | Experiment design + recording + analysis [10]                                   | Trial available [10]      |
| Reading assessment       | Lexplore                   |     | Service + Hardware  | AI-based reading assessment [11]                                                | Quote-based [12]          |
| Accessibility device     | OrCam Read 3               |     | Hardware            | Reads printed/digital text aloud [13]                                           | \$116/month option [13]   |
| Adaptive interface       | Microsoft Immersive Reader |     | Built-in feature    | Line Focus; Text Spacing; read-aloud [14]                                       | Included [14]             |
| E-reading customization  | Amazon Kindle settings     |     | Device feature      | Font size and appearance adjustments [15]                                       | Included with device [15] |

### 1.7.4 Gaps and Opportunities

#### Architecture Gap: Experimentability

The project proposal highlights that tightly coupled prototypes hinder systematic comparison of interventions and decision strategies. Empirical findings show that intervention design materially affects recovery and user preference [2, 16].

**Opportunity:** Design the adaptation engine with hot-swappable intervention modules and auditable logging to enable:

- Systematic A/B testing,
- Replay-based evaluation,
- Real-time override capabilities,
- Transparent experimental comparison.

### **1.7.5 Conclusion**

The reviewed literature and market landscape collectively demonstrate that adaptive reading systems require:

1. Robust gaze-processing pipelines,
2. Carefully designed intervention mechanisms,
3. Context-preservation metrics,
4. Modular, experiment-ready architectures.



## **2 Methodology**

- 2.1 Research and Engineering Approach**
- 2.2 Requirements Elicitation**
- 2.3 Development Process**
- 2.4 Evaluation Strategy**
- 2.5 Tools and Technologies**
- 2.6 Ethical Considerations**





## **3 Requirements and Analysis**

- 3.1 Stakeholders and User Roles**
- 3.2 Use Cases and Scenarios**
- 3.3 Functional Requirements**
- 3.4 Non-Functional Requirements**
- 3.5 Constraints**
- 3.6 Summary**



## **4 System Design and Architecture**

### **4.1 Architectural Overview**

### **4.2 Monorepo Structure and Rationale**

### **4.3 Component Design**

#### **4.3.1 Gaze Acquisition and Preprocessing**

#### **4.3.2 Event Extraction (Fixations, Saccades, Regressions)**

#### **4.3.3 Reading-State Classification**

#### **4.3.4 Intervention Engine**

#### **4.3.5 Logging and Replay Subsystem**

### **4.4 Data Model and Interfaces**

### **4.5 Design Decisions and Alternatives**

### **4.6 Summary**



## **5 Implementation**

### **5.1 Monorepo Layout**

### **5.2 Build, CI and Developer Workflow**

### **5.3 Selected Implementation Highlights**

#### **5.3.1 Real-Time Gaze Pipeline**

#### **5.3.2 Hot-Swappable Intervention Modules**

#### **5.3.3 Session Recording and Replay**

### **5.4 Cross-Cutting Concerns**

### **5.5 Challenges and Resolutions**

### **5.6 Summary**



## **6 Evaluation**

- 6.1 Evaluation Setup**
- 6.2 Functional Verification**
- 6.3 Performance Evaluation**
- 6.4 Experimentability and Developer Experience**
- 6.5 Case Study / Proof of Concept**
- 6.6 Threats to Validity**
- 6.7 Summary**





## **7 Discussion**

**7.1 Achievements vs. Problem Statement**

**7.2 Comparison with Related Work**

**7.3 Limitations**

**7.4 Lessons Learned**

**7.5 Future Work**



## **8 Conclusion**



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# A Declaration of Use of Generative AI

## A.1 Statement of Authorship and Responsibility

We, the authors of this thesis, take full and sole responsibility for the content of this report. All decisions regarding scope, claims, results, and interpretations are ours. Generative artificial intelligence (GAI) tools listed below were used as assistive instruments only; they are not co-authors of this work.

## A.2 Tools Used

| Tool | Version / Date | Purpose |
|------|----------------|---------|
|------|----------------|---------|

Table A.1: Generative AI tools used during the preparation of this thesis.

## A.3 Where and How GAI Was Used

### A.3.1 Writing Assistance

### A.3.2 Code Assistance

### A.3.3 Literature Search and Synthesis

### A.3.4 Figures, Tables, or Data

## A.4 What GAI Was Not Used For

- Formulation of the problem statement and research questions.
- Selection and reading of all cited sources.
- Design decisions for the system architecture.
- Evaluation methodology and interpretation of results.
- Conclusions and discussion of limitations.

## A.5 Prompt and Output Retention

Representative prompts and outputs are retained by the authors and can be made available to the supervisor or examiners upon request, in accordance with DTU guidance on documenting GAI use.

## A.6 Citations of GAI Output

Where text, code, or figures derived from GAI output appear in the body of the thesis, they are cited inline using the format below.

Anthropic. *Claude* (Opus 4.7) [Large Language Model]. <https://claude.ai>. Accessed YYYY-MM-DD.



Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

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